

Section 4.4

1) $y' = X^2 - X - 2, y'' = 2X - 1$

→ y increasing at $(-\infty, -1) \cup (2, \infty)$, decreasing at $(-1, 2)$

∴ Local min: $X=2$, Local max: $X=1$

→ Inflection point: $0 = y'' = 2X - 1 \rightarrow X = \frac{1}{2}$

→ Concave up ($y'' > 0$) → $(\frac{1}{2}, \infty)$

→ Concave down ($y'' < 0$) → $(-\infty, \frac{1}{2})$

4.) $y = \frac{9}{14}X^{7/3} - \frac{9}{2}X^{1/3} \rightarrow y' = \frac{3}{2}X^{4/3} - \frac{3}{2}X^{-2/3} \rightarrow y'' = 2X^{1/3} + X^{-5/3} = X^{-5/3}(2X^2 + 1)$

→ $y'' > 0$ (Concave up) when $X > 0$ and $y'' < 0$ (Concave down) when $X < 0 \rightarrow$ Inflection point at $(0, 0)$.

→ $0 = y' = \frac{3}{2}X^{-2/3}(X^2 - 1) \therefore X = 1, -1$ as local extremes

At $X=1, y'' > 0 \rightarrow$ Local minimum, at $X=-1, y'' < 0 \rightarrow$ Local maximum

17.) $y' = 4X^3 - 4X \stackrel{\text{set}}{=} 0 \therefore X = -1, 0, 1$

Slope $\leftarrow \begin{array}{cccc} < 0 & > 0 & < 0 & > 0 \end{array}$

∴ Local min: $(-1, -1), (1, -1)$, Local max: $(0, 0)$

$y'' = 12X^2 - 4 \stackrel{\text{set}}{=} 0 \rightarrow X = -\frac{1}{3}, \frac{1}{3}$

Concave $\leftarrow \begin{array}{ccc} \text{up} & \text{down} & \text{up} \\ & -1/3 & 1/3 \end{array}$

∴ Inflection point: $(-\frac{1}{3}, -\frac{5}{9}), (\frac{1}{3}, -\frac{5}{9})$

23.) $y' = 1 + \cos X \stackrel{\text{set}}{=} 0 \rightarrow X = \pi + 2k\pi \ (k \in \mathbb{Z})$

($y' > 0$ for all points except $X = \pi + 2k\pi$)

$y'' = -\sin X = 0 \rightarrow X = k\pi \ (k \in \mathbb{Z})$

Concave $\leftarrow \begin{array}{ccc} < 0 & > 0 & < 0 & > 0 \\ & 2k\pi - \pi & 2k\pi & 2k\pi + \pi \end{array}$

32.) $y' = \frac{-(X+2)}{(2X+1)^2 \sqrt{1-X^2}} \therefore$ Decreasing at all points in domain

$(X \in [-1, 1] - \{-\frac{1}{2}\})$

$y'' = \frac{-4X^3 - 12X^2 + 7}{(2X+1)^3 (1-X^2)^{3/2}}$

∴ concave up at $(-0.92, 0.5) \cup (0.5, 0.69)$, Concave down at $(-1, -0.92) \cup (0.69, 1)$

Inflection point at $X \approx -0.92, X \approx 0.69$



46.) $y = |X^2 - 2X|$

$X \leq 0$ or $X \geq 2$ $0 < X < 2$

$y = X^2 - 2X$

$2X - X^2$

$y' = 2X - 2$

$2 - 2X$

→ Local min: $X=0, X=2$

$< 0; X \leq 0$

$> 0; X < 1$

Local max: $X=1$

$> 0; X \geq 2$

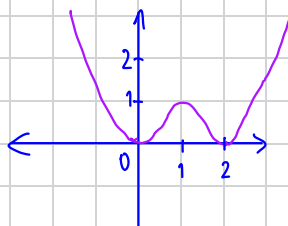
$< 0; X > 1$

$y'' = 2$

-2

Concave up

Concave down





76) $y = \frac{x^2 - 49}{x^2 + 5x - 14} = \frac{(x+7)(x-7)}{(x+7)(x-2)} = \frac{x-7}{x-2} = 1 - \frac{5}{x-2} ; x \neq -7, 2$

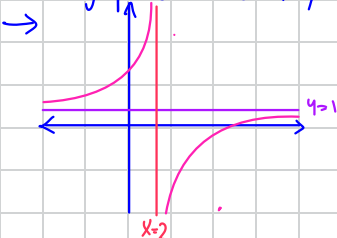
1) Domain: All x except $x = -7, 2$, No symmetry

2) $y' = \frac{5}{(x-2)^2}$, $y'' = -\frac{10}{(x-2)^3}$

3) No absolute extrema, increasing on $(-\infty, 2) \cup (2, \infty)$ (hole at $x = -7$)

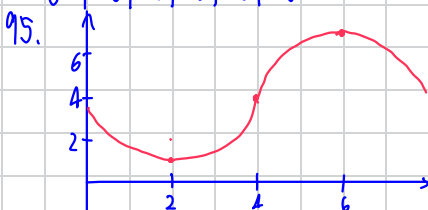
4) No inflection points (concave up on $(2, \infty)$ and concave down on $(-\infty, 2)$)

5) Asymptotes: $y = 1$, $x = 2$



93.

	P	Q	R	S	T
y'	< 0	> 0	> 0	$= 0$	< 0
y''	> 0	$= 0$	< 0	< 0	< 0



98. a) Away: $t = 1.5$ s to $t = 4$ s, $t = 10$ s to $t = 12$ s, $t = 13.5$ s to $t = 16$ s
Toward: $t = 0$ s to $t = 1.5$ s, $t = 4$ s to $t = 10$ s, $t = 12$ s to $t = 13.5$ s

b) Approximately $t = 4$ s, $t = 12$ s ($f'(t) = 0$)

c) Approximately $t = 1.8$ s, $t = 6$ s, $t = 8$ s, $t = 11$ s, $t = 13.5$ s

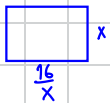
d) Acceleration positive: $t = 0$ s to $t = 1.5$ s, $t = 6$ s to $t = 8$ s, $t = 11$ s to $t = 13.5$ s
negative: $t = 1.8$ s to $t = 6$ s, $t = 8$ s to $t = 11$ s, $t = 13.5$ s to $t = 16$ s

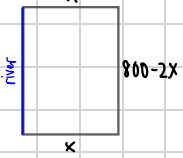
99. The marginal cost is $\frac{dc}{dx}$ which changes from decreasing to increasing when $\frac{d^2c}{dx^2} = 0$, or the inflection point of the curve which is approximately 60 thousand units.

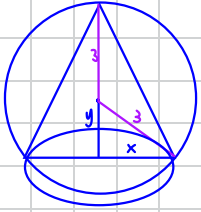
104. It means that the second derivative of the graph will always be either positive or negative. In other words, it is always either concave up or concave down.

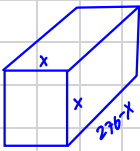
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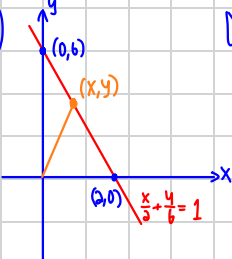
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1.  Perimeter $P(x) = 2\left(x + \frac{16}{x}\right) = 2x + \frac{32}{x}$; $0 < x < 16$
 $\therefore P'(x) = 2 - \frac{32}{x^2}$, $P''(x) = \frac{64}{x^3} > 0$ for all $x \in (0, 16)$
 Want: Minimum $P(x) \rightarrow P'(x) = 0$
 $2 - \frac{32}{x^2} = 0 \rightarrow x = 4$, $\frac{16}{x} = \frac{16}{4} = 4$
 $\therefore P(4) = 2(4) + \frac{32}{4} = 16$ for the square of side 4.

7.  Area $A(x) = x(800 - 2x) = 800x - 2x^2$ Domain: $[0, 400]$
 Finding C.P. $\rightarrow 0 = A' = 800 - 4x$ $\therefore x = 200$, $800 - 2x = 400$
 $\therefore A(200) = 800(200) - 2(200)^2 = 80000 \text{ m}^2$

12.  As $3^2 = y^2 + x^2 \rightarrow x^2 = 9 - y^2$
 $\therefore$ Volume of cone $V(y) = \frac{1}{3} \pi x^2 (y + 3)$ ($y \in (0, 3)$)
 $= \frac{1}{3} \pi (9 - y^2)(y + 3) = \frac{\pi}{3} (-y^3 - 3y^2 + 9y + 27)$
 Finding critical points: $0 = V'(y) = \frac{\pi}{3} (-3y^2 - 6y + 9)$
 $\therefore 0 = y^2 + 2y - 3 \rightarrow y = -3, 1$ $\therefore x = 2\sqrt{2}$
 $\therefore V(1) = \frac{1}{3} \pi (9 - 1^2)(1 + 3) = \frac{32}{3} \pi$
 $V(0) = 9\pi$, $V(3) = 0 \rightarrow V(1) = \frac{32}{3} \pi$ is the maximum volume.

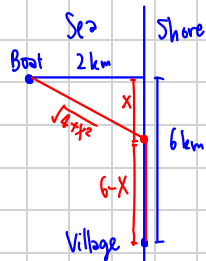
20. a)  Volume $V(x) = x^2(276 - 4x) = 276x^2 - 4x^3$; $x \in (0, 138)$
 Finding critical points: $0 = V'(x) = 552x - 12x^2$
 $\therefore x = 0, 46$
 $V''(46) = 552 - 24(46) < 0 \rightarrow x = 46 \text{ cm}$ is the local maximum
 $\therefore \text{length} = 276 - 2(46) = 184 \text{ cm}$

28.)  Distance from point (x, y) to origin $D(x) = \sqrt{x^2 + y^2}$
 As $\frac{x}{a} + \frac{y}{b} = 1$, $y = b(1 - \frac{x}{a}) = b - \frac{bx}{a}$
 $\therefore D(x) = \sqrt{x^2 + (b - \frac{bx}{a})^2} = \sqrt{x^2 + b^2 - \frac{2b^2x}{a} + \frac{b^2x^2}{a^2}}$
 $= \sqrt{(1 + \frac{b^2}{a^2})x^2 - \frac{2b^2}{a}x + b^2}$
 Critical points: $0 = D'(x) = \frac{(2 + \frac{2b^2}{a^2})x - \frac{2b^2}{a}}{2\sqrt{(1 + \frac{b^2}{a^2})x^2 - \frac{2b^2}{a}x + b^2}}$
 $\therefore x = \frac{\frac{2b^2}{a}}{2 + \frac{2b^2}{a^2}} = \frac{b^2 a}{a^2 + b^2} \rightarrow y = \frac{a^2 b}{a^2 + b^2}$
 $D(\frac{ab^2}{a^2 + b^2}) = \sqrt{\frac{a^4 b^4}{(a^2 + b^2)^2} + \frac{a^4 b^4}{(a^2 + b^2)^2}} = \sqrt{\frac{2a^4 b^4}{(a^2 + b^2)^2}} = \frac{ab}{\sqrt{a^2 + b^2}}$
 $D(0) = b$, $D(a) = a \leftrightarrow D(0) > D(\frac{ab^2}{a^2 + b^2})$ and $D(a) = D(\frac{ab^2}{a^2 + b^2})$
 $\leftrightarrow x = \frac{ab^2}{a^2 + b^2}$ is the local minimum
 $\leftrightarrow (\frac{ab^2}{a^2 + b^2}, \frac{a^2 b}{a^2 + b^2})$ is the point on the line $\frac{x}{a} + \frac{y}{b} = 1$ that is closest to origin.

Section 4.5 (Continue)

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38.



Let Jane arrives at x km down the shoreline from the point nearest the boat.

$\therefore$ Total distance rowing = $\sqrt{4+x^2}$, total distance walking = $6-x$

$\rightarrow$ Total time travelling $t(x) = \frac{\sqrt{4+x^2}}{2} + \frac{6-x}{5}$

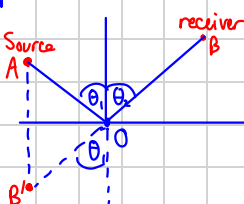
Critical points: $0 = t'(x) = \frac{x}{2\sqrt{4+x^2}} - \frac{1}{5}$

$$5x = 2\sqrt{4+x^2}$$

$$95x^2 = 4x^2 + 16 \rightarrow x = \frac{4}{\sqrt{21}}$$

We consider only from $x=0$ to $x=6$ (Endpoints) and when we compare $t(x)$ at critical point and endpoints, we get $t(0) = \frac{11}{5}$, $t(\frac{4}{\sqrt{21}}) = \frac{6+\sqrt{21}}{5}$, $t(6) = \sqrt{10}$ and $t(\frac{4}{\sqrt{21}})$ is the smallest among the three, hence Jane should land her boat $\frac{4}{\sqrt{21}}$ km down the shoreline from the point nearest her boat.

48.



To minimize the distance light has to travel, $B'O'B$ must be a straight line. Hence, $\theta_1 = \theta_2$.

49. From $v = k\alpha x - kx^2$, we get $v' = \frac{dv}{dx} = k\alpha - 2kx$.

Let $v' = 0$, we have $2kx = k\alpha$

$$x = \frac{\alpha}{2}$$

$v'' = -2k$ meaning this critical point is the local maximum. $\therefore v$ at $x = \frac{\alpha}{2}$ equals $k\alpha(\frac{\alpha}{2}) - k(\frac{\alpha}{2})^2 = \frac{k\alpha^2}{4}$

51. Profit $P(x) = nx - nc = n(x-c) = a + b(100-x)(x-c) = a - 100bc - bx^2 + b(c+100)x$

At critical points, we get $0 = P'(x) = -2bx + b(c+100)$, $P''(x) = -2b < 0 \rightarrow$ Critical points $\Rightarrow$ maximum

$$2bx = b(c+100) \rightarrow x = \frac{c+100}{2} \text{ will bring the maximum profit.}$$

54. Let $c(x)$ is the cost of producing x items; $x > 0$ and $\frac{c(x)}{x}$ being the average cost of producing x items assumed to be differentiable. If the average cost can be minimized, it will be at a production level at which

$$0 = \frac{d}{dx} \left(\frac{c(x)}{x} \right) = \frac{xc'(x) - c(x)}{x^2} \text{ or } xc'(x) - c(x), \text{ leading to } \frac{c(x)}{x} = c'(x) \text{ which concludes the proof.}$$

67. a) The distance from point $(\frac{3}{2}, 0)$ to $(x, \sqrt{x})$ is $D(x) = \sqrt{(x-\frac{3}{2})^2 + (\sqrt{x}-0)^2} = \sqrt{x^2 - 2x + \frac{9}{4}}$

To find the critical point, we set $0 = D'(x) = \frac{x-1}{\sqrt{x^2-2x+\frac{9}{4}}}$ or $x=1$ $\xleftarrow{\text{slope} = +}$ $\xrightarrow{+}$

$$\therefore D(1) = \sqrt{(1-\frac{3}{2})^2 + (1-0)^2} = \frac{\sqrt{5}}{2} \text{ will be the shortest distance.}$$

Section 4.6

1. $f(x) = x^2 + x - 1$, $f'(x) = 2x + 1$

$$x_0 = -1 \rightarrow x_1 = -1 - \frac{(-1)^2 + (-1) - 1}{2(-1) + 1} = -2$$

$$x_2 = -2 - \frac{(-2)^2 + (-2) - 1}{2(-2) + 1} = -\frac{5}{3}$$

$$x_0 = 1 \rightarrow x_1 = 1 - \frac{1^2 + 1 - 1}{2(1) + 1} = \frac{2}{3}$$

$$\rightarrow x_2 = \frac{2}{3} - \frac{(\frac{2}{3})^2 + (\frac{2}{3}) - 1}{2(\frac{2}{3}) + 1} = \frac{13}{21}$$

Section 4.6 (Continue)

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3. $f(x) = x^4 + x - 3$, $f'(x) = 4x^3 + 1$

$$X_0 = -1 \rightarrow X_1 = -1 - \frac{(-1)^4 - 1 - 3}{4(-1)^3 + 1} = -2$$

$$X_2 = -2 - \frac{(-2)^4 + (-2) - 3}{4(-2)^3 + 1} = -\frac{51}{31}$$

$$X_0 = 1 \rightarrow X_1 = 1 - \frac{1^4 + 1 - 3}{4(1)^3 + 1} = \frac{6}{5}$$

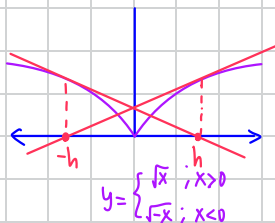
$$X_2 = \frac{6}{5} - \frac{(\frac{6}{5})^4 + \frac{6}{5} - 3}{4(\frac{6}{5})^3 + 1} \approx 1.165$$

7. From $X_1 = X_0 - \frac{f(X_0)}{f'(X_0)}$, if we know that X_0 is the root of $f(X_0)$ and $f'(X_0)$ is defined but not equal to 0, we get that $X_1 = X_0 - 0 = X_0$ and $X_n = X_{n-1} = \dots = X_1$ for all $n \in \mathbb{N}$.

9. $f = \begin{cases} \frac{1}{2\sqrt{x}} & ; x > 0 \\ -\frac{1}{2\sqrt{x}} & ; x < 0 \end{cases}$

If $X_0 = h > 0$, $X_1 = h - \frac{\frac{1}{2\sqrt{h}}}{\frac{1}{2\sqrt{h}}} = h - 2h = -h$

If $X_0 = -h < 0$, $X_1 = -h - \frac{-\frac{1}{2\sqrt{h}}}{-\frac{1}{2\sqrt{h}}} = h$



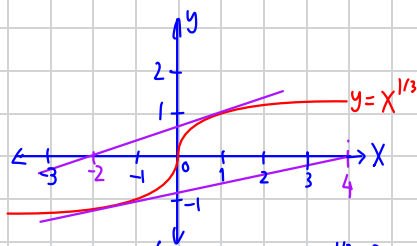
10. $f(x) = x^{1/3}$, $f'(x) = \frac{1}{3}x^{-2/3}$

$$X_0 = 1 \rightarrow |X_1| = \left| 1 - \frac{1^{1/3}}{\frac{1}{3} \cdot 1^{-2/3}} \right| = |-2| = 2 = 2^1$$

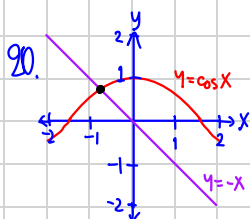
$$|X_2| = \left| -2 - \frac{(-2)^{1/3}}{\frac{1}{3} \cdot (-2)^{-2/3}} \right| = |-2 + 6| = 4 = 2^2$$

$$|X_3| = \left| 4 - \frac{4^{1/3}}{\frac{1}{3} \cdot 4^{-2/3}} \right| = |4 - 12| = 8 = 2^3$$

$$|X_4| = \left| -8 - \frac{(-8)^{1/3}}{\frac{1}{3} \cdot (-8)^{-2/3}} \right| = |-8 + 24| = 16 = 2^4$$



$\rightarrow |X_n| = 2^n$ and $|X_n| \rightarrow \infty$ when $n \rightarrow \infty$, since $X_0 = 1$ is not close to $X_r = 0$ (which is root of $X^{1/3} = 0$) enough



From the graph, we let $f(x) = \cos x + x \rightarrow X_{n+1} = X_n - \frac{X_n + \cos X_n}{1 - \sin X_n}$

Let $X_0 = -0.7$, we get $X_1 = -0.73944$, $X_2 = -0.73908$ $\therefore$ at $X \approx 0.74$, we have $\cos X = X$

